

Paramedic Specialist

Chemical Response Data Sheet

NaOH 2018 1.0
Chemical Name:

Sodium Hydroxide (NaOH) 50%
Solid and Solution

Identifying Data:

 UN # 1823 (solid)
UN# 1824 (solution)

CAS # 1310-73-2

Classification: Corrosive

NaOH (Solid)

Placard

NFPA Rating

WHMIS 2015 Symbols

Physical Properties:

Colour	Clear/colourless liquid White Flakey solids
Vapor Density	>1
Solubility	Completely soluble
Boiling point	145C
PH	> 12
Freezing Point	14C
Specific Gravity	2.13 (Heavier then water)
Vapour Pressure (total)	1.5 mmHg at 20C. ***

*** May pose inhalation hazard if heated or reacting

NaOH (Solution)

General Description:

Colour less liquid or white/granular solid. **Very strong base.** Contact with eyes may result in loss of vision. Severe burns on skin, slippery to touch (*Sapnification*). Not an inhalation hazard unless heated or reacting with other chemicals or metals. Tranported and used in various concentrations. Common ingrediant in many house hold cleaning products.

Commonly Found Loactions: (Various concentrations)

- | | | |
|-------------------------|-----------------------------------|-----------------------|
| • Transport | • Textile Manufaturing | • Dehorning of cattle |
| • Pulp /Paper Mills | • Drain cleaner (Drano) | • Aluminum etching |
| • Plastic manufaturing | • Soap and detergent manufaturing | |
| • Clandestine Drug labs | | |

Work Safe BC Permissible Limits

TWA N/A STEL/Ceiling 2mg/m3

IDLH 10mg/M3 (SCBA must be worn at or above this level)

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Paramedic Specialist Procedures to Consider

- Do initial Size-up of incident. (# of patients, LOC, other hazards present)
- Evacuate immediate area and deny entry to anyone not wearing appropriate PPE.
- Have patients perform self Decon if capable. (Shirts not to be pulled over face).
- **DO NOT DELAY PATIENT DECONTAMINATION.** (See Decon section below)
- Eyes Take precedence.
- Concentrations greater than 40% may generate sufficient heat to ignite contaminated combustibles when reacting with water. Reactions with water may bring NaOH above boiling point causing dangerous sporadic eruptions and splash hazards.



(see also)

TDG Reference: Guide 154

Emergency Decontamination:

- Have patient self-Decon with direction as needed from safe distance. Assist removing clothing if needed with appropriate PPE to prevent secondary contamination and injury.
- Use water for decontamination. Large volume/low pressure. Tepid if Immediately available.
- Eyes take precedence.
- Goal is to begin Decon within 1 minute of contact if possible (Time = Tissue).
- Ensure patient is fully decontaminated. Do not forget skinfolds, armpits, groin, buttocks and feet.
- Use PH paper to test for contaminant remaining.
- Do not cause hypothermia. Focus on symptomatic areas (pain, visible burns).
- See Pre-Hospital Care below for further information.

Primary route of exposure: Eyes/Skin/Inhalation

Primary Targets of Toxicity: Skin/Eyes/Lungs

- Sodium Hydroxide is an extremely corrosive Alkali. Skin contact results in deep structure burns due to liquefaction necrosis. (Deep burns without a protective coagulum (scar) forming.)
- Slippery to touch. This is the process of saponification (turning into soap), or the fats of bodies cells being dissolved by the caustic allowing further penetration of the caustic into the tissues.
- Eyes: Sodium Hydroxide is an extreme eye hazard. Delayed effects of up to 72 hours have been seen. Permanent vision loss may occur.
- Toxic by ingestion, rare in Industry (intentional only). May cause upper airway burns/swelling and GI tract burns and bleeding.

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- Inhalation of Sodium Hydroxide fumes/mists or vapour may result in upper airway stridor, dyspnea (shortness of breath) and Pulmonary Edema.

Pre- Hospital Care:

- Remove victim from hazardous area utilizing appropriate PPE for hazard.
- **Immediately** remove all contaminated clothing from victim(s). Begin flushing with water while removing clothing if immediately available. **Medium volume/low pressure only. Remove any nozzles from hoses.**
- **Eyes take priority.** Flush immediately as per training and continue to flush en route to Hospital or until medical aid arrives. Flushing should continue until hospital PH test of surface of eyes proves all contaminant has been removed and some recommendations are for 2 hours after.
- **Immediate Decontamination of skin required. DO NOT DELAY.** Flush with water as per training for chemical burns. Ensure all areas of body are thoroughly flushed and decontaminated. (*Head, eyes, hair, armpits, groin, genitalia, all skin folds etc.*) Serious lasting burns have resulted by neglecting these areas during decontamination of corrosives and oxidizers.
- Treat patient following support of ABC's as per training.
- Any Cardiovascular (Tachycardia, Bradycardia, hypotension and/or Neurologic symptoms confusion/DLOC/seizures) may be secondary due to hypoxia or fluid loss due to extensive burns.
- In the event of severe inhalation exposure patient may present with non-Cardiogenic pulmonary edema (Bilateral crackles heard with auscultation at lung bases, may be delayed finding.)
- Exposed patients with underlying lung disease (Asthma, COPD) may become symptomatic at lower concentrations than healthy individuals. Utilize their prescribed inhalers as required.
- Contact BC Poison Control at 1-800-567-8911 or 604-682-5050 for Physician advisement on patient care.

Fire and Reactivity:

- Non-combustible. Utilize high volumes of water when extinguishing adjacent combustibles involved. CO2 or dry chemical if available. Do not use Halogen type extinguishing agents.
- Reacts with water. Concentrations greater than 40% may generate sufficient heat to ignite contaminated combustibles when reacting with water. Reactions with water may bring NaOH above boiling point causing dangerous sporadic eruptions and splash hazards.
- Generates significant heat when mixed with acids.
- Sodium Hydroxide will react with aluminum and other metals and produce hydrogen gas.
- Incompatible and reacts with Flammable liquids; organic halogens; metals such as aluminum, tin, zinc (galvanized), magnesium, bronze, brass, and copper.
- Reacts with various reducing sugars to produce Carbon Monoxide.
- Numerous less common incompatibles.

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References and Links:

WISER (Wireless Information System for Emergency Responders)

AHLS Provider Manual/4th edition.

WorkSafe BC/Exposure Limits

CDC Toxnet.